

Cooperation for environmental sustainability – a common pool resource virtual experience

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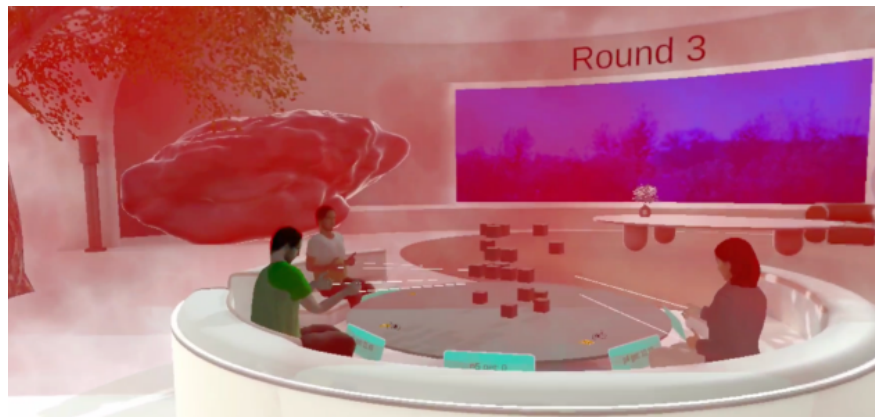


Figure 1: VR game for sustainable cooperation

ABSTRACT

The presented application is a cooperative game based on a theoretical model of human behavior in the Tragedy of the Commons [1], of which climate change is one of the prime examples. In this common pool resource (CPR) dilemma, climate is the common resource which bears the burden of humans extracting non-renewable energy sources. Maximizing individual resource exploitation is at odds with the group's collective profit, which is affected by the overall resource depletion. Sustainable use refers to the balance between short-term individual gains and the long term common gains while not damaging the resource.

Developed at the University of Warsaw, and implemented in VR by Kiin (formerly Virtual Bodyworks), as part of the EU-funded GuestXR project, led by Dr. Mel Slater from the University of Barcelona, the game places groups of 3-5 players in a shared environment where they make critical decisions about harvesting energy cubes—symbolic units of resource extraction. Players are guided by the avatar of an AI agent, GuestXR, who explains the rules

and offers intermittent feedback to encourage cooperation and warn against the dangers of overuse.

A mathematical model of the CPR dilemma was developed and used as the engine of the game as well as in the simulations needed to develop the Guest XR interventions. In line with economic theory, each player in our game aims to maximize their utility through resource harvesting. The dilemma of choosing the proper harvesting rate that maximizes individual total gain without external coordination poses a challenge. Based on the literature review, we identified key features that can alter the group dynamic and promote sustainable usage of the resources and included them in the model:

1. Awareness of the state of the resources and consequences of their use, where low awareness results in non-optimality of harvesting, usually over-harvesting. The awareness of a low state of the resources is enabled by the visual display of the number of energy blocks remaining, by the state of the environment visible in the window from green rich vegetation to brown leafless trees, and the state of the tree growing in the room (changes color and loses leaves). At a very high level of depletion of the common resources, red smoke spreads through the virtual room. Additionally, the Guest verbally warns the players about their overuse of the resource.
2. Time perspective, where concentration on the short-time gains results in a greedy behavior and overuse of the resource, while the long-time perspective reads of sustainable harvesting of the resources. The players see on the screen their predicted gains if everyone in the group harvests the energy cubes at the current rate. This variable is influenced by the Guest XR, by varying the time perspective for which the gains are calculated.

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3. Perceived similarity with the other players that increases the sustainability of player's behaviors and cooperative orientations. This variable is manipulated by the coordination exercise (or lack of it) in the beginning of the game, and by the form of the verbal instructions (targeted at individual player or the group as a whole).

Primary results obtained within our CPR framework are promising as the model allows to re-create dynamics obtained with human participants in the laboratory settings and in the field (varying from recreation of viscous cycles to responsible harvesting). In line with the economic theory, our results are sensitive to the environmental capacity, starting point of the environment, number of players who need to coordinate, and their internal parameters.

The gameplay

The scenario centers on cryptocurrency mining, an energy-consuming process that significantly contributes to environmental degradation due to high electricity usage and carbon emissions. As a currently trendy phenomenon, cryptocurrency mining serves as an ecologically valid metaphor for climate overexploitation and was thus incorporated into the game storyline. After connecting to the application, players enter the space which resembles a peculiar room with a window overlooking a natural landscape, a tree planted inside and a grey rock resembling an unspecified mineral resource. Players sit around the shared resource pile - a stack of cube-shaped blocks symbolizing units of energy for cryptocurrency mining. The avatar of the AI agent, GuestXR, who explains the game rules, is also present in the room at the beginning of the game. Even after she disappears, players can still hear her vocal interventions (e.g. encouraging cooperation, balanced resource mining, and forewarning about the consequences of over-extraction).

The game is played in rounds. In each round, participants harvest energy cubes by pointing at them and firing their motion controllers. Individual income is depicted through progressively changing symbols of wealth (coins, a motorbike, a car, and a mansion). Instructions state that the person with the highest income at the end of the game will receive a monetary bonus. However, the resource is not infinite and although it regrows, the regrowth rate is unknown. Overharvesting results in lack of sustainability (both for players, and for the environment). Under-harvesting results in inability to satisfy the basic level of virtual 'income' that players are required to sustain throughout the game.

Technical details

The XR application developed by Kiin using Unity3D for standalone Meta Quest devices is available on the Meta Horizon Store upon request. It provides a platform for both single and multi-user XR experiences. In multi-user experiences, remote participants are embodied in life-size, full-body avatars within a shared XR environment, enabling natural interaction through body and hand movements, facial expressions, and voice. The interactive logic and dynamics of the game were implemented in Python, leveraging Kiin's Python SDK to interface with the shared environment. This SDK facilitates interaction with virtual objects, control over avatar behavior, sound playback, and modification of auditory, visual, and haptic elements. User actions, such as pressing the controller's trigger button or through hand gestures while pointing at a virtual object, can trigger corresponding events in the connected Python client; for example, initiating a mining sound, making an object appear or disappear and updating the game state accordingly. An example is the gradual appearance of red or yellow smoke around participants or a change in a tree's color to reflect a less healthy state of the environment in response to depleting game resources caused by participants' actions.

Closest related work

A similar VR application, also dedicated to CPR, is an extraction game currently under development at BarcelonaTech, Universitat Politècnica de Catalunya [2], where groups of 3 participants take on

the roles of lumberjacks in a forest. The goal of the planned experiments is to compare the effects of the environment with and without immersive socio-affective cues on user sustainability decisions. Like in our approach, the decisions on how many trees (resource units) to harvest are made based on information provided in discrete, synchronized rounds; the resource replenishment/degradation is visualized; participants receive wealth tokens for the manufactured wood.

Novelty of our project

The novelty of our approach lies in several factors, with the most important being the introduction of a virtual assistant, the GuestXR, a virtual instructor powered by AI technology. GuestXR learns from players' behavior and introduces interventions to facilitate sustainable decisions. The AI agent acts as a guide, helping participants succeed in the game and resolve the CPR dilemma. The goal of its interventions goes beyond merely shaping attitudes relating to sustainable behavior. They equip participants with essential competencies related to sustainability, such as the ability to accurately perceive the state of shared resources, predict the future consequences of their actions on other players and the environment, and understand how to balance environmental stability with personal gain. While players aim to maximize individual profits, the ultimate success in the game depends on their ability to use shared resources sustainably. GuestXR actively assists in this process, teaching players how to collaborate and anticipate long-term effects of their actions.

The project also integrates a theoretical foundation with a mathematical model of behavior in the Tragedy of the Commons scenario. This model describes the dynamics of the shared environment and the impact of players' actions. The interventions delivered by GuestXR are based on critical variables identified in CPR literature. Specifically, the model includes three factors that increase the likelihood of sustainable behavior: awareness of the state of the environment, time perspective, and perceived similarity to other players. GuestXR's interventions are directly targeted at these key aspects.

From a technological perspective, the project introduces innovative elements in the VR space. The online platform features immersive special effects, such as smoke and falling leaves, that visually represent the outcomes of players' and GuestXR's interventions. Moreover, as a multiplayer social game in the metaverse, it transcends geographical barriers. Participants from around the globe, with diverse skills and perspectives on climate change, can come together to engage in our experience. In summary, our project represents a novel approach to addressing the CPR dilemma by combining cutting-edge AI, immersive VR technology, and a robust theoretical framework to teach players sustainable behaviors in an engaging way.

Index Terms: Common pool resource, environmental sustainability, virtual reality, game.

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